

Diagram and Data Dictionary

Internal database structure of the Data Management Portal

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Tables: 17

Columns: 179

1. How to read this document

Every table is created by `migrate()`, in `backend/db.php`, which runs on each request and is idempotent: it creates what is missing and leaves the rest untouched. The real name carries the configured prefix -- with the default `gdt`, `usuarios` becomes `gdt_usuarios`.

Types are shown in a neutral form because the portal supports four engines. The mapping is: primary key with auto-increment, long text and timestamp are resolved per engine at creation time.

There are no FOREIGN KEYS. Integrity between related tables (a user and their photo, a change and its objects) is enforced in application code, which keeps the schema portable across MySQL, PostgreSQL, SQL Server and SQLite.

2. Overview

Accounts, session and security

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Governance modules

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Portal configuration

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3. Dictionary

usuarios

Portal accounts, with role, visual preferences and security controls.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
username	VARCHAR(50) NOT NULL UNIQUE	Login name, unique in the portal.
nome_completo	VARCHAR(120)	Name shown in the interface.
password_hash	VARCHAR(255) NOT NULL	Password as bcrypt. Never leaves through the API.
role	VARCHAR(20) NOT NULL DEFAULT 'leitura'	Role: leitura, escrita, admin or master.
criado_em	date and time	Creation date and time.
atualizado_em	date and time	Date and time of the last change.
modulos_permitidos	long text (TEXT / NVARCHAR(MAX))	Allowed modules, comma separated.
email	VARCHAR(150)	E-mail, used to send the verification code.
mfa_ativo	INT NOT NULL DEFAULT 0	1 when two-step verification is on.
cor_tema	VARCHAR(20) NOT NULL DEFAULT 'padrao'	Color palette chosen by the user.
estilo_side	VARCHAR(20) NOT NULL DEFAULT 'claro'	Sidebar appearance (light or dark).
ativo	INT NOT NULL DEFAULT 1	0 disables the account without erasing history.
must_change_password	INT NOT NULL DEFAULT 0	1 forces a password change on next sign-in.
tokens_valid_after	date and time	Revocation cutoff: tokens issued before this are rejected.

usuarios_foto

Profile photo, in a separate table so the image is not read on every authenticated request.

Column	Type	Description
usuario_id	INT NOT NULL PRIMARY KEY	User who owns the photo. Primary key, without auto-increment.
foto	very long text (MEDIUMTEXT on MySQL)	Image as a data URI (base64), already resized to 256x256 by the browser.
atualizado_em	date and time	Date and time of the last change.

tentativas_login

Failed login counters, the basis for throttling per user+IP and per IP.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
usuario	VARCHAR(150) NOT NULL	Module field.
ip	VARCHAR(45) NOT NULL	Source IP address.
tentativas	INT NOT NULL DEFAULT 0	Code verification attempts (max 5).
primeira_tentativa	date and time	Module field.
ultima_tentativa	date and time	Module field.
bloqueado_ate	date and time	Module field.

tokens_revogados

Identifiers (jti) of tokens invalidated before expiry, by logout or password change.

Column	Type	Description
jti	VARCHAR(64) NOT NULL PRIMARY KEY	Identifier of the revoked token.
expira_em	date and time	Code validity (10 minutes).

mfa_codigos

Two-step verification codes, stored only as an HMAC.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
usuario	VARCHAR(150) NOT NULL	Module field.
token	VARCHAR(64) NOT NULL	Verification session token, tied to the code.
codigo_hash	VARCHAR(64) NOT NULL	HMAC of "token code". The plain code is never stored.
tentativas	INT NOT NULL DEFAULT 0	Code verification attempts (max 5).
expira_em	date and time	Code validity (10 minutes).
criado_em	date and time	Creation date and time.
reenvios	INT NOT NULL DEFAULT 0	Resends already made in this login session (max 5).
ultimo_reenvio	DATETIME	Time of the last resend, basis for the 60s cooldown.

auditoria_log

Audit trail: who changed what, when and from which IP.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
tabela	VARCHAR(60) NOT NULL	Module field.
registro_id	INT	Module field.
acao	VARCHAR(40) NOT NULL	Module field.
usuario	VARCHAR(150)	Module field.
dados_antes	long text (TEXT / NVARCHAR(MAX))	Module field.
dados_depois	long text (TEXT / NVARCHAR(MAX))	Module field.
ip	VARCHAR(45)	Source IP address.
criado_em	date and time	Creation date and time.

acessos

Inventory of access granted to databases and systems.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
data	DATE	Module field.
usuario	VARCHAR(150)	Module field.
tipo	VARCHAR(50)	Classification, from the Cadastro lists.
objeto	VARCHAR(200)	Module field.
nivel	VARCHAR(100)	Module field.
justificativa	long text (TEXT / NVARCHAR(MAX))	Module field.
solicitante	VARCHAR(150)	Module field.
aprovador	VARCHAR(150)	Module field.
revisao	DATE	Module field.
status	VARCHAR(30) DEFAULT 'Ativo'	Current status, from the Cadastro lists.
obs	long text (TEXT / NVARCHAR(MAX))	Additional notes.
criado_em	date and time	Creation date and time.
atualizado_em	date and time	Date and time of the last change.
servidor	VARCHAR(150)	Server where the item resides.
criado_por	VARCHAR(150)	Login of whoever created the record, set by the server.

mudancas

Log of changes to data structures.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
codigo	VARCHAR(30)	Module field.
data	DATE	Module field.
ambiente	VARCHAR(30)	Module field.
tipo	VARCHAR(50)	Classification, from the Cadastro lists.
descricao	long text (TEXT / NVARCHAR(MAX))	Free description of the record.
script	VARCHAR(200)	Module field.
rollback	long text (TEXT / NVARCHAR(MAX))	Module field.
solicitante	VARCHAR(150)	Module field.
aprovador	VARCHAR(150)	Module field.
status	VARCHAR(30)	Current status, from the Cadastro lists.
resultado	long text (TEXT / NVARCHAR(MAX))	Module field.
criado_em	date and time	Creation date and time.
atualizado_em	date and time	Date and time of the last change.
criado_por	VARCHAR(150)	Login of whoever created the record, set by the server.

mudancas_objetos

Objects affected by each change (child of mudancas).

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
mudanca_id	INT NOT NULL	Module field.
nome	VARCHAR(200) NOT NULL	Record name.
tipo	VARCHAR(50)	Classification, from the Cadastro lists.
criado_em	date and time	Creation date and time.

backup_politicas

Backup policy per database: window, retention and destination.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
banco	VARCHAR(150)	Related database.
criticidade	VARCHAR(30)	Criticality level, from the Cadastro lists.
tipo	VARCHAR(50)	Classification, from the Cadastro lists.
frequencia	VARCHAR(100)	Routine periodicity (daily, weekly, monthly).
horario	VARCHAR(50)	Frequency interval -- how often it repeats.
retencao	VARCHAR(100)	Module field.
local	VARCHAR(250)	Module field.
rpo	VARCHAR(50)	Module field.
rto	VARCHAR(50)	Module field.
responsavel	VARCHAR(150)	Person or team responsible.
criado_em	date and time	Creation date and time.
atualizado_em	date and time	Date and time of the last change.
criado_por	VARCHAR(150)	Login of whoever created the record, set by the server.

restore_testes

Restore tests performed, with duration and result.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
data	DATE	Module field.
banco	VARCHAR(150)	Related database.
backup	VARCHAR(150)	Module field.
tempo	VARCHAR(50)	Module field.
resultado	VARCHAR(30)	Module field.
por	VARCHAR(150)	Module field.
obs	long text (TEXT / NVARCHAR(MAX))	Additional notes.
criado_em	date and time	Creation date and time.
atualizado_em	date and time	Date and time of the last change.
criado_por	VARCHAR(150)	Login of whoever created the record, set by the server.

dicionario_dados

Business dictionary of the tables and columns of the managed databases.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
banco	VARCHAR(150)	Related database.
schema_nome	VARCHAR(150)	Module field.
tabela	VARCHAR(150)	Module field.
coluna	VARCHAR(150)	Module field.
tipo_dado	VARCHAR(100)	Module field.
permite_nulo	VARCHAR(10)	Module field.
descricao	long text (TEXT / NVARCHAR(MAX))	Free description of the record.
classificacao	VARCHAR(50)	Module field.
origem	VARCHAR(150)	Module field.
obs	long text (TEXT / NVARCHAR(MAX))	Additional notes.
criado_em	date and time	Creation date and time.
atualizado_em	date and time	Date and time of the last change.
servidor	VARCHAR(150)	Server where the item resides.
criado_por	VARCHAR(150)	Login of whoever created the record, set by the server.

integracoes

Integrations between systems, with source, target and execution window.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
nome	VARCHAR(200)	Record name.
origem	VARCHAR(150)	Module field.
ip_origem	VARCHAR(45)	Module field.
destino	VARCHAR(150)	Module field.
ip_destino	VARCHAR(45)	Module field.
tipo	VARCHAR(60)	Classification, from the Cadastro lists.
direcao	VARCHAR(30)	Module field.
mecanismo	VARCHAR(150)	Module field.
frequencia	VARCHAR(30)	Routine periodicity (daily, weekly, monthly).
dados_trafegados	long text (TEXT / NVARCHAR(MAX))	Module field.
classificacao	VARCHAR(50)	Module field.
criticidade	VARCHAR(30)	Criticality level, from the Cadastro lists.
resp_tecnico	VARCHAR(150)	Module field.
resp_negocio	VARCHAR(150)	Module field.
ambiente	VARCHAR(30)	Module field.
status	VARCHAR(30)	Current status, from the Cadastro lists.
ultima_revisao	DATE	Module field.
obs	long text (TEXT / NVARCHAR(MAX))	Additional notes.
criado_em	date and time	Creation date and time.
atualizado_em	date and time	Date and time of the last change.
criado_por	VARCHAR(150)	Login of whoever created the record, set by the server.

jobs

Scheduled routines: command, steps, frequency, schedules and owner.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
nome	VARCHAR(200)	Record name.
tipo	VARCHAR(60)	Classification, from the Cadastro lists.
servidor	VARCHAR(150)	Server where the item resides.
banco	VARCHAR(150)	Related database.
descricao	long text (TEXT / NVARCHAR(MAX))	Free description of the record.
comando	long text (TEXT / NVARCHAR(MAX))	Job steps as JSON: step name and what it runs.
frequencia	VARCHAR(60)	Routine periodicity (daily, weekly, monthly).
horario	VARCHAR(60)	Frequency interval -- how often it repeats.
agendamentos	long text (TEXT / NVARCHAR(MAX))	List of windows (start and end) as JSON.
criticidade	VARCHAR(30)	Criticality level, from the Cadastro lists.
status	VARCHAR(30)	Current status, from the Cadastro lists.
responsavel	VARCHAR(150)	Person or team responsible.
obs	long text (TEXT / NVARCHAR(MAX))	Additional notes.
criado_por	VARCHAR(150)	Login of whoever created the record, set by the server.
criado_em	date and time	Creation date and time.
atualizado_em	date and time	Date and time of the last change.

config_tipos

Editable lists used in the forms (type, environment, criticality, status).

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
categoria	VARCHAR(40) NOT NULL	Module field.
nome	VARCHAR(120) NOT NULL	Record name.
criado_em	date and time	Creation date and time.

config_email

SMTP server settings; the password is stored with AES-256-GCM.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
host	VARCHAR(150)	Module field.
porta	INT	Module field.
seguranca	VARCHAR(10) DEFAULT 'tls'	Module field.
usuario	VARCHAR(150)	Module field.
senha_cifrada	long text (TEXT / NVARCHAR(MAX))	Module field.
remetente_nome	VARCHAR(120)	Module field.
remetente_email	VARCHAR(150)	Module field.
testado_ok	INT NOT NULL DEFAULT 0	1 after a successful test send.
atualizado_em	date and time	Date and time of the last change.

config_projeto

Portal identity: name, logo and session inactivity timeout.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
nome_projeto	VARCHAR(160)	Module field.
logo_data	very long text (MEDIUMTEXT on MySQL)	Portal logo as a data URI. MEDIUMTEXT on MySQL because of the 64KB TEXT limit.
atualizado_em	date and time	Date and time of the last change.
timeout_inatividade_min	INT NOT NULL DEFAULT 30	Minutes of inactivity before the session ends.

4. Relationships

From	To	Rule
usuarios_foto.usuario_id	usuarios.id	One photo per user. Removed together with the user.
mudancas_objetos.mudanca_id	mudancas.id	Objects affected by a change. Removed with the parent record.
auditoria_log.usuario	usuarios.username	By login, not by id: the trail survives the account being deleted.
mfa_codigos.usuario	usuarios.username	Verification session in progress.
tentativas_login.usuario	usuarios.username	Also records logins that do not exist, to avoid revealing valid accounts.